

Baseline Analysis Brief

EV-SDG — *Synthetic Data Generator for Electric Vehicle Charging Sessions: Modeling and Evaluation Using Real-World Data*

Lahariya, Benoit & Develder — *Energies* 2020, 13(16):4211 (open access; journal extension of the ACM e-Energy 2020 paper). Reference baseline in our study.

1. What the paper does

EV-SDG addresses the scarcity of openly available EV charging data by fitting a generative model to a large real-world dataset (ElaadNL, Netherlands) and sampling realistic synthetic charging sessions. Each real session is a tuple of start date-time, connection (plug-in) time, requested energy, and charge point. The generator is built from three modules:

- **Arrival Model (AM):** EV arrivals as a stochastic process — either an inter-arrival-time (IAT) exponential process or an arrival-count (AC) process. The arrival rate λ varies with time-of-day / day-type / season and is estimated by mean, polynomial/loess regression, or Poisson / negative-binomial regression.
- **Connection-time model (MM_c):** a Gaussian Mixture Model for plug-in duration, *conditional on the arrival time slot*. Departure = arrival + connection time.
- **Energy model (MM_e):** a Gaussian Mixture Model for requested energy, again conditional on the arrival time slot.

Before fitting, sessions and charge points (“poles”) are grouped into behaviourally similar clusters (DBSCAN); the models are then fit at the aggregate / cluster level. Generation over a horizon samples arrival instants from the AM and then draws connection time and energy from the conditional GMMs. The tool is open-sourced with trained models. Evaluation compares synthetic vs. real distributions of arrival, connection time and energy, showing faithful reproduction of *aggregate* patterns.

2. EV-SDG as a baseline in our study (combined dataset, 140 users, 28,159 sessions)

We re-ran a clean EV-SDG (IAT / mean arrival model) on the exact combined set of selected profiles and scored population-level divergences against the real data (identical binning). Two independent runs agreed to three decimals.

Variable	KL (real■synth)	JS	KS	W_1
Arrival hour	0.040	0.011	0.098	1.26
Plug-in duration	1.706	0.039	0.097	2.93
Energy	2.853	0.043	0.123	2.93
Mean KL	1.533			

EV-SDG reproduces the **aggregate arrival shape** well (KL 0.040) but is weak on **duration** (1.71) and **energy** (2.85).

3. Why EV-SDG under-performs our method

- **Population/cluster-level pooling — no per-user modelling.** EV-SDG fits shared (cluster-level) distributions across heterogeneous users, so each user's distinct duration and energy signature is averaged out. The pooled marginals are smoothed, which is exactly why duration (1.71) and energy (2.85) diverge so much.
- **Arrival-conditioned GMMs cannot capture multimodal, heavy-tailed behaviour.** Duration and (especially) energy are multimodal and heavy-tailed at the individual level; a GMM conditioned only on the arrival slot, with few components, cannot represent them — energy is the worst case (2.85).
- **Independence / no temporal structure.** EV-SDG treats sessions as independent draws (Poisson arrivals + independent conditional samples). It has no notion of session-to-session transitions or the *idle gap* between sessions — a variable it does not model at all.
- **Its single strength is aggregate arrival timing.** The Poisson rate per time slot captures the arrival marginal well (KL 0.040, close to ours), but that is the only variable where pooling is sufficient.

- **Designed for large aggregated public charging.** Tuned to ElaadNL (thousands of poles); on smaller residential cohorts its clustering pools behaviour and, empirically, degenerates — it cannot even run on our individual Korea (30) / Pecan (10) cohorts.

4. Head-to-head with the proposed per-user semi-Markov scheme

Our method fits a **per-user semi-Markov chain** (session-as-state encoding, per-user transition matrix and sojourn / idle-gap distributions) with per-bin BIC-selected families (LogNormal / Exponential / Weibull / Truncated GMM). Population-level KL (nats) from the framework’s L4 validation:

KL (nats)	EV-SDG (combined)	Ours Norway	Ours Korea	Ours Pecan
Arrival hour	0.040	0.010	0.008	0.034
Plug-in duration	1.706	0.019	0.018	0.021
Energy	2.853	0.003	0.014	0.015
Idle gap	— (not modelled)	0.003	0.047	0.025
Mean (A,D,E)	1.533	0.011	0.013	0.024

Our per-user method keeps KL in the **0.003–0.047** range on every variable and dataset (all rated “excellent”), i.e. roughly **60–150× lower** mean KL than EV-SDG’s pooled model, and it additionally models the idle gap that EV-SDG omits.

Note on fairness: EV-SDG’s figures are on the pooled 140-user combined set; our figures are per-dataset (the like-for-like combined run via `run_smc.jl` combined is pending, but the per-dataset results establish the 1–2 order-of-magnitude gap on duration and energy). EV-SDG’s clustering cannot run on the small cohorts individually, which is itself evidence for a per-user approach.

5. Takeaway

EV-SDG is a solid, widely-cited *parametric, population-level* baseline whose Poisson arrival process matches aggregate arrival timing. It under-performs our scheme because it pools heterogeneous users and conditions only on arrival time, so it cannot reproduce the per-user, multimodal duration/energy distributions or the sequential idle-gap structure that the per-user semi-Markov chain captures explicitly. The gap is largest exactly where per-user heterogeneity matters most — energy (2.85 vs. ≤ 0.015) and duration (1.71 vs. ≤ 0.021).